How to give read and write permissions using Unix commands. A very simple example is to create a text file with a C program, then use chmod.

Create a file called permission.c

```
#include <stdio.h>
int main()
{
    FILE *fp;
    fp = fopen("sample.txt", "w");
    if (fp == NULL)
    {
        printf("Cannot open file\n");
        return 1;
    }
    fprintf(fp, "Hello, this is a test file.\n");
    fclose(fp);
    printf("File created successfully.\n");
   return 0;
}
```

Compile

```
gcc permission.c -o permission
```

Run

```
./permission
```

Output
File created successfully.

A file called sample.txt is created.

Check the current permissions

ls -l sample.txt

You may see

-rw-r--r-- 1 user user 31 Aug 17 08:20 sample.txt

The owner has: Read + Write

Group has: Read

Others have: Read

This is permission 644.

Give Read and Write permission to everyone

chmod 666 sample.txt

Check ls -l sample.txt

You should see

-rw-rw-rw- 1 user user ... sample.txt

Instead of numbers, you can directly give:

chmod u+rw sample.txt

chmod o+rw sample.txt

chmod a+rw sample.txt

Task of the Day

1) Give Read and Write only to the owner
2) Give Read and Write to the owner and group
3) Only owner can read and write
4) Owner read/write, everyone else read-only
5) Owner can do everything, others can only read and execute
6) Owner and group can read/write
7) Only owner can execute

```
shashank@SHASHANK: ~/per

shashank@SHASHANK:~$ mkdir  permission.c
shashank@SHASHANK:~$ cd  permission.c
shashank@SHASHANK:~/permission.c$ nano  permission.c
shashank@SHASHANK:~/permission.c$ gcc permission.c -o permission.c
gcc: fatal error: input file ‘permission.c’ is the same as output file
compilation terminated.
shashank@SHASHANK:~/permission.c$ gcc permission.c -o permission
shashank@SHASHANK:~/permission.c$ ./permission
File created successfully.
shashank@SHASHANK:~/permission.c$ ls -l sample.txt
-rw-r--r-- 1 shashank shashank 28 Aug 17 15:57 sample.txt
shashank@SHASHANK:~/permission.c$ chmod 600 sample.txt
shashank@SHASHANK:~/permission.c$ ls -l sample.txt
-rw------- 1 shashank shashank 28 Aug 17 15:57 sample.txt
shashank@SHASHANK:~/permission.c$ chmod 660 sample.txt
shashank@SHASHANK:~/permission.c$ ls -l sample.txt
-rw-rw---- 1 shashank shashank 28 Aug 17 15:57 sample.txt
shashank@SHASHANK:~/permission.c$ chmod 600 sample.txt
shashank@SHASHANK:~/permission.c$ ls -l sample.txt
-rw------- 1 shashank shashank 28 Aug 17 15:57 sample.txt
shashank@SHASHANK:~/permission.c$ chmod 644 sample.txt
shashank@SHASHANK:~/permission.c$ ls -l sample.txt
-rw-r--r-- 1 shashank shashank 28 Aug 17 15:57 sample.txt
shashank@SHASHANK:~/permission.c$ chmod 755 sample.txt
shashank@SHASHANK:~/permission.c$ ls -l sample.txt
-rwxr-xr-x 1 shashank shashank 28 Aug 17 15:57 sample.txt
shashank@SHASHANK:~/permission.c$ chmod 660 sample.txt
shashank@SHASHANK:~/permission.c$ ls -l sample.txt
-rw-rw---- 1 shashank shashank 28 Aug 17 15:57 sample.txt
shashank@SHASHANK:~/permission.c$ chmod 100 sample.txt
shashank@SHASHANK:~/permission.c$ ls -l sample.txt
---x------ 1 shashank shashank 28 Aug 17 15:57 sample.txt
shashank@SHASHANK:~/permission.c$ |
```

Fill in the permission numbers for the following:

8) Owner: read + write; Group: nothing; Others: nothing →600___________

9) Owner: read + write; Group: read; Others: read →644

____

10) Owner: read + write + execute; Group: read + execute; Others: read + execute → 755_______

11) Owner: read + write; Group: read + write; Others:

____

nothing →660____

12) Everyone: read only →444___